RESEARCH AND EDUCATIONAL USE ONLY — NOT A MEDICAL DEVICE

Pharmacogenomic analysis summary

sample	NA12273
analysed	2026-11-06T12:28:47.254230Z
printed	2026-11-06
drugs checked	2

This is not a clinical report. It was produced by a final-year student project for research and education. It has not been clinically validated, and no qualified clinical expert has reviewed the explanatory text on this page.

Recommendation text on this page is quoted from published CPIC guidance via PharmCAT. Automated source checks are not human review: they cannot show that every statement here is supported, or that any of it is correct for the person this sample came from.

Do not change a medicine on the strength of this page. Take it to a qualified healthcare professional.

Results

Ordered by consequence, not alphabetically.

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codeine **Ineffective**

gene	CYP2D6
diplotype	*4/*4
phenotype	PM
evidence level	Unknown

CYP2D6 supplied by Cyrius, not called by PharmaGuard.

codeine: CYP2D6 *4/*4 (Poor Metabolizer) — Ineffective.

CPIC GUIDELINE — QUOTED EXACTLY

Avoid codeine use because of possibility of diminished analgesia. If opioid use is warranted, consider a non-tramadol opioid.

CPIC strength of recommendation: Strong. Population: general. Implications: CYP2D6: Greatly reduced morphine formation leading to diminished analgesia.

— Annotation of CPIC Guideline for codeine and CYP2D6 (via PharmCAT) [label rule: avoid_for_therapeutic_failure]

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clopidogrel **Unknown**

gene	Unknown
diplotype	Unknown
phenotype	Unknown
evidence level	Unknown

clopidogrel: no pharmacogenomic result (Unknown).

APPLICATION LIMITATION — NOT A CPIC QUOTE

Required genomic evidence is incomplete for CYP2C19. Available partner findings are partial evidence only; no complete drug-level PGx recommendation is made.

Annotation of CPIC Guideline for clopidogrel and CYP2C19 — PharmCAT (no matching CPIC entry)

Unknown results (1)

An Unknown is a declined answer, not a missing one. Only the reasons about the file itself — missing positions or unusable genotypes — are something the sample provider can act on.

clopidogrel **Your file is missing required positions**

Some of the positions needed to identify this gene carry no genotype in your file. That is not a small gap: a change whose key position is absent is invisible, so the genotype would read as normal and a reduced-function result could be reported as safe. The system declines rather than risk that.

What would fix it: Upload a VCF that reports ALL positions, including those matching the reference — a clinical pharmacogenomic panel, or data covering all your genes, produced with every position reported. A variants-only file cannot work here.

Input coverage

How many of the positions each gene needs carried a usable genotype in the uploaded file. A genotype caller reads an absent position as matching the reference, so missing data can produce confident results rather than uncertain ones — which is why the count is printed whether or not it passed, and why a gene below its minimum is reported as Unknown.

gene	usable	required	minimum	sufficient
CYP2C19	16	35	100%	NO
CYP2C9	17	88	100%	NO
CYP2D6	67	157	100%	NO
CYP4F2	0	20	100%	NO
DPYD	31	83	20%	NO
NUDT15	5	20	80%	NO
SLCO1B1	20	35	100%	NO
TPMT	9	45	80%	NO
VKORC1	0	1	100%	NO

Not resolved from this file: CYP2C19, CYP2C9, CYP2D6, CYP4F2, DPYD, NUDT15, SLCO1B1, TPMT, VKORC1. Any drug keyed to these genes was reported as Unknown rather than assumed normal.

Where this guidance came from

PharmCAT	3.4.0
CPIC data version	2026-07-13-11-40
CPIC table SHA-256	5010bb813ed6aaacf504c689e1f74924ddc7c58383d5e35604074d299345d6a8
allele definitions	2026-07-13-11-40

definitions SHA-256	7044c87682de18646a74b890739e2cc145ef0a3a21718a17b371dc013c7afe8f
explanations generated	24 Jul 2026
source	CPIC guidelines, retrieved via PharmCAT

Guidance reflects what CPIC published when this data was captured. CPIC revises its guidelines. A weekly repository check compares this build's CPIC data with CPIC's current release and flags differences for review; an analysis does not check at the time it runs.